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BSCSSE – AN43 **11/1/2026**

TA 3 - DOCUMENTATION

Smart Pantry Meal Planner Using Rule-Based and Case-Based Reasoning

The Smart Pantry Meal Planner is an intelligent system that recommends meals based on the ingredients available in a household pantry. The knowledge base stores information about the available ingredients, cooking time, expiration status, number of ingredients, and type of meal. The system can recommend meals such as chicken fried rice, egg and cheese sandwich, stir-fried vegetable noodles, and vegetable omelet. It also considers whether an ingredient is close to expiring so that it can be prioritized and used before it goes to waste. This system can be useful for households that want to plan meals conveniently while maximizing their available ingredients. It can also help reduce food waste and save time when deciding what to cook.

Knowledge Representation

Knowledge Representation is implemented using a Python dictionary called facts. It stores the important information about the user's pantry and meal preferences.

The facts include:

- Available ingredients
- Cooking time
- Whether an ingredient is expiring soon
- Number of available ingredients
- Type of meal

The program represents household knowledge in a structured format so that the reasoning systems can examine the information and produce recommendations.

Rule-Based Reasoning Explanation

The Rule-Based Reasoning system uses IF–THEN logic. Each rule checks a specific condition in the knowledge base. For example, if chicken and rice are available and the cooking time is short or medium, the system recommends chicken fried rice. If the user has an ingredient that will expire soon, the system recommends prioritizing that ingredient. The inference engine

checks all the rules one by one and displays every recommendation that applies to the current pantry situation.

Case-Based Reasoning Explanation

The Case-Based Reasoning system uses previous meal-planning cases stored in a case base. Each case contains a set of pantry features and a previously recommended meal. When a new pantry situation is entered, the system compares it with the old cases. It counts how many features match between the new situation and each previous case. The case with the highest similarity score is retrieved, and its meal recommendation is reused.

Similarity Method

The similarity method compares the ingredients in the new pantry situation with the ingredients in each previous case. Each matching feature receives one point. For example, if both the new problem and an old case contain chicken, rice, and egg, those matching features increase the similarity score. The case with the highest score is considered the most similar case. This method is simple and appropriate for the program because the meal recommendations are based mainly on the presence or absence of specific ingredients.

Adaptation Logic

The adaptation process happens during the revise step of the CBR cycle. After the system retrieves and displays a meal recommendation, the user is asked whether the recommendation should be revised. If the user selects “yes,” they can enter a different preferred meal. If the user selects “no,” the original recommendation is retained. The revised or unchanged recommendation is then stored as part of a new case. This allows the system to remember the result and expand its case base for future recommendations.

SCREENSHOTS

Attempt 1:

```
SMART PANTRY MEAL PLANNER

Available ingredients:
Chicken, rice, egg, vegetables, noodles, bread, cheese

Enter the ingredients you have, separated by commas: chicken, rice, egg
Cooking time: short, medium, or long: long
Do you have an ingredient that will expire soon? yes/no: yes

KNOWLEDGE BASE
Ingredients: ['chicken', 'rice', 'egg']
Cooking time: long
Has expiring ingredient: True
Number of ingredients: 3
Meal type: household meal

RULE-BASED REASONING RESULTS
- Prioritize using the ingredient that will expire soon.

CASE-BASED REASONING RESULTS
Current pantry: ['chicken', 'rice', 'egg']
Most similar past case: {'chicken': True, 'rice': True, 'egg': True, 'vegetables': False, 'noodles': False, 'bread': False,
'cheese': False}
Similarity score: 3 out of 7
Reused meal recommendation: Chicken fried rice

Do you want to revise the recommendation? yes/no: no
The original recommendation will be retained.

FINAL MEAL RECOMMENDATION: Chicken fried rice
The new meal case has been added to the case base.
Total cases in case base: 5
```

Attempt 2:

```
SMART PANTRY MEAL PLANNER

Available ingredients:
Chicken, rice, egg, vegetables, noodles, bread, cheese

Enter the ingredients you have, separated by commas: vegetables, noodles, egg
Cooking time: short, medium, or long: short
Do you have an ingredient that will expire soon? yes/no: no

KNOWLEDGE BASE
Ingredients: ['vegetables', 'noodles', 'egg']
Cooking time: short
Has expiring ingredient: False
Number of ingredients: 3
Meal type: household meal

RULE-BASED REASONING RESULTS
- Stir-fried vegetable noodles are a suitable meal.
- A vegetable omelet is a suitable meal.

CASE-BASED REASONING RESULTS
Current pantry: ['vegetables', 'noodles', 'egg']
Most similar past case: {'chicken': False, 'rice': False, 'egg': False, 'vegetables': True, 'noodles': True, 'bread': False,
 'cheese': False}
Similarity score: 2 out of 7
Reused meal recommendation: Stir-fried vegetable noodles

Do you want to revise the recommendation? yes/no: no
The original recommendation will be retained.

FINAL MEAL RECOMMENDATION: Stir-fried vegetable noodles
The new meal case has been added to the case base.
Total cases in case base: 5
```

REFLECTION

Rule-Based Reasoning was easier to implement because it only required clearly defined IF–THEN statements. Each rule directly connected a pantry condition to a recommended action or meal. Case-Based Reasoning required more steps because the program needed to store previous cases, compare features, calculate similarity scores, retrieve a solution, and retain the new case. However, CBR was more flexible because it could use previous meal-planning experiences even when the new pantry situation was not exactly the same as an existing rule. RBR is more suitable when the conditions and results are predictable, such as recommending fried rice when chicken and rice are available. CBR is useful when there are many possible combinations of ingredients and solutions can be improved through previous experiences. The CBR system also becomes more useful over time because each new meal recommendation can be added to the case base. In real life, RBR can be used in automatic appliance controls, cooking instructions, and household safety systems. CBR can be used in recipe recommendation systems, customer support, medical consultation, and personalized household planning.

CONCLUSION

The Smart Pantry Meal Planner demonstrates how knowledge representation allows information to be organized in a form that a computer can process. The Rule-Based Reasoning component provides recommendations using predefined conditions. The Case-Based Reasoning component solves a new pantry situation by comparing it with previous meal-planning cases. Both systems are useful, but they approach problems differently. RBR depends on fixed rules, while CBR learns from stored experiences. Together, they show how intelligent systems can support simple household decision-making.